

Ex. No.: 11b)

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LRU

Aim:

To write a c program to implement LRU page replacement algorithm.

Algorithm:

- 1: Start the process
- 2: Declare the size
- 3: Get the number of pages to be inserted
- 4: Get the value
- 5: Declare counter and stack
- 6: Select the least recently used page by counter value
- 7: Stack them according the selection.
- 8: Display the values
- 9: Stop the process

Program Code: #include

```
<stdio.h> int findLRU(int time[],
int n) {      int i, min = time[0],
pos = 0;      for (i = 1; i < n; ++i)
{          if (time[i] < min) {
min = time[i];      pos = i;
}
}
return pos;
} int main() { int frames[10], pages[30],
counter[10];  int i, j, k, pos, max, faults = 0,
time = 0;    int n, f;      printf("Enter
number of frames: ");      scanf("%d", &f);
printf("Enter number of pages: ");
scanf("%d", &n);
printf("Enter reference string: ");
for (i = 0; i < n; ++i)
scanf("%d", &pages[i]);      for (i = 0;
i < f; ++i) {
frames[i] = -1;
counter[i] = 0;
```

```

    }
    printf("\n");    for (i = 0; i < n; ++i) {        int
flag1 = 0, flag2 = 0;        for (j = 0; j < f; ++j) {
if (frames[j] == pages[i]) {            time++;
counter[j] = time; // Update recent use time        flag1
= flag2 = 1;            break;
        } } if (flag1 == 0) { for (j
        = 0; j < f; ++j) {

if (frames[j] == -1) {
time++;            faults++;
frames[j] = pages[i];
counter[j] = time;            flag2
= 1;            break;
        }
    } } if (flag2 == 0)
    {
        pos
= findLRU(counter, f);
time++;    faults++;
frames[pos] = pages[i];
counter[pos] = time;
    }
    // Display current frame state
for (k = 0; k < f; ++k) {        if
(frames[k] != -1)
printf("%d ", frames[k]);
else        printf("-1 ");
    }
    printf("\n");
    }
    printf("\nTotal Page Faults = %d\n", faults);
    return 0;
}

```

OUTPUT:

```
Enter number of frames: 3
Enter number of pages: 10
Enter reference string: 3
2
6
8
3
4
1
2
2
6

3 -1 -1
3 2 -1
3 2 6
8 2 6
8 3 6
8 3 4
1 3 4
1 2 4
1 2 4
1 2 6

Total Page Faults = 9
```

RESULT:

Hence, page faults that occur using LRU page replacement technique has been found.